CS & IT ENGINEERING

Compiler Design

Syntax Directed Translation

DPP 01 Discussion Notes



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[MCQ]

#Q. Synthesized attributes can be easily simulated using

Children Bottom up approach

A LL grammar

B LR grammar

C Ambiguous grammar

D None of these

[MCQ]

#Q. Consider the following translation rules for the grammar G:

S → a {print “A”} A
A → b {print “C”}B
A → ∈ {print ”C”}
B → e {print “B”}A
B → ∈ {print “C”}
C → c {print “A”}

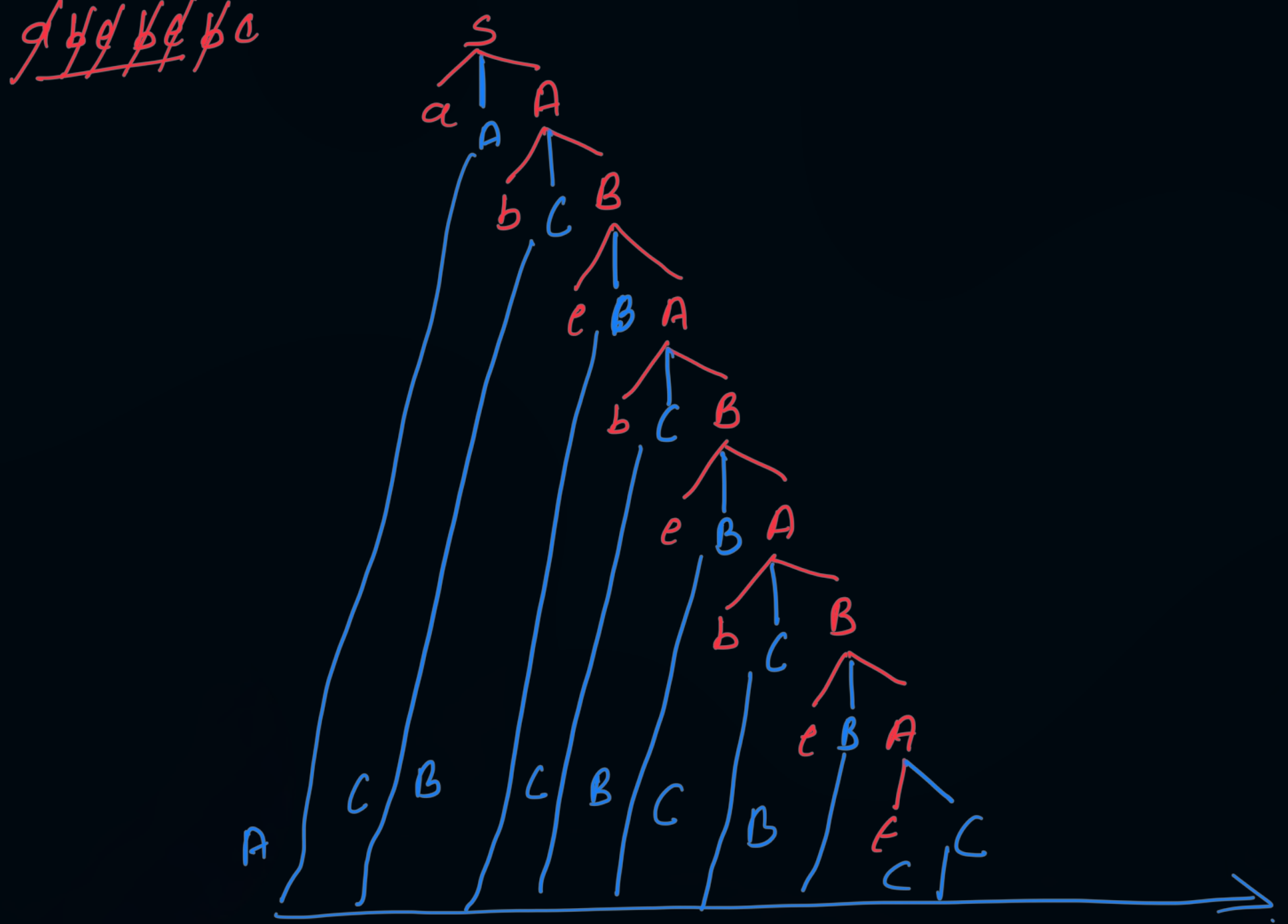
What will be the output for the input string "abebebe" you top-down parser?

A ACBCCBAC

B ACCBCCBC

C ACBCCBCC

D ACBCBCBC


S
a A
A
b C B
e B A
b C B
e B A
b C B
e B A
F C
C
A C B C B C B C

[MCQ]

#Q. Consider the following attribute grammar:

$A \rightarrow BA'$	$A' \cdot b = A \cdot a$
	$A \cdot a = A' \cdot b$
$A_1' \rightarrow +BA_2'$	$A_2' \cdot b = A_1' \cdot b + B \cdot a$
	$A_1' \cdot a = A_2' \cdot a$

Which of the following is true?

Synthesized → Children (a)

Inherited → Parent (b)

A. Both a and b are inherited attributed. ✗

B. Both a and b are synthesized attributed. ✗

C. a is inherited, b is synthesized ✗

D. b is inherited, a is synthesized ✓

[MSQ]

#Q. Consider the following grammar:

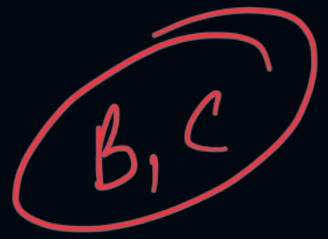
$E \rightarrow E + T \mid T$

$T \rightarrow T/F \mid F$

$F \rightarrow F * A \mid A$

$A \rightarrow id$

Which one of the following is/are true?







A / have higher precedence than * False

B *have higher precedence than + True

C + have lower precedence than / True

D *, +, / all have same precedence. False

[MCQ]

#Q. A shift reduce parser perform action specified within process immediately after reduction to the corresponding rule of grammar.

S → abV {print '11'}

S → cc {print '22'}

V → Sd {print '33'}

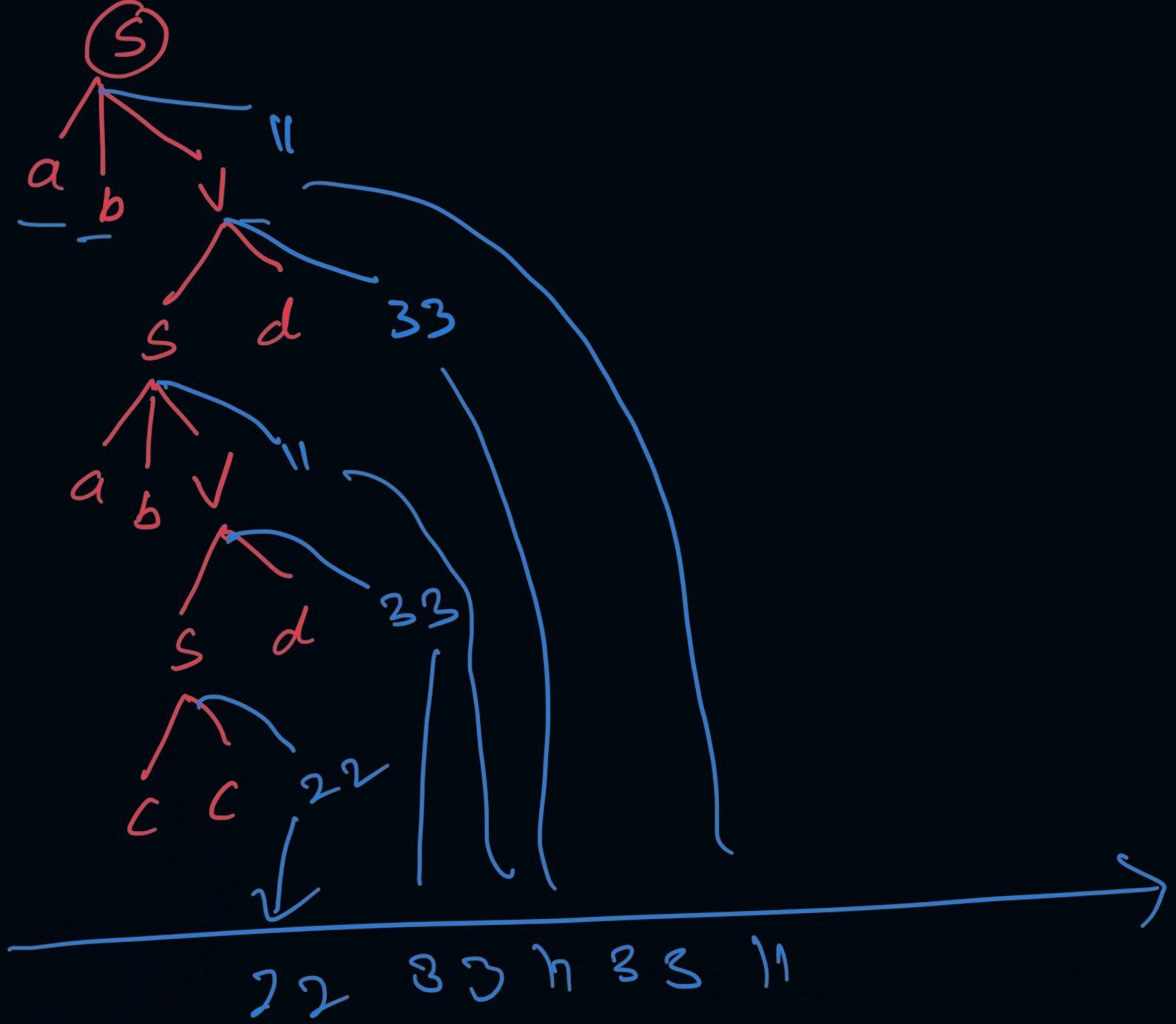
What is the translation of "ababccdd" using the SDT scheme described by above rules?

A 2233113311

B 1133113322

C 2211331122

D 1122113322


S
a
b
V
11
S
d
33
a
b
V
11
S
d
33
C
C
22
22
33
44
55
11

[MCQ]

#Q. Consider the following transition rules:

$A \rightarrow BC$

$C \rightarrow +BC \mid A + \mid \in$

$B \rightarrow DB$ {print '+'} $\mid \in$

$D \rightarrow$ id {print number value}

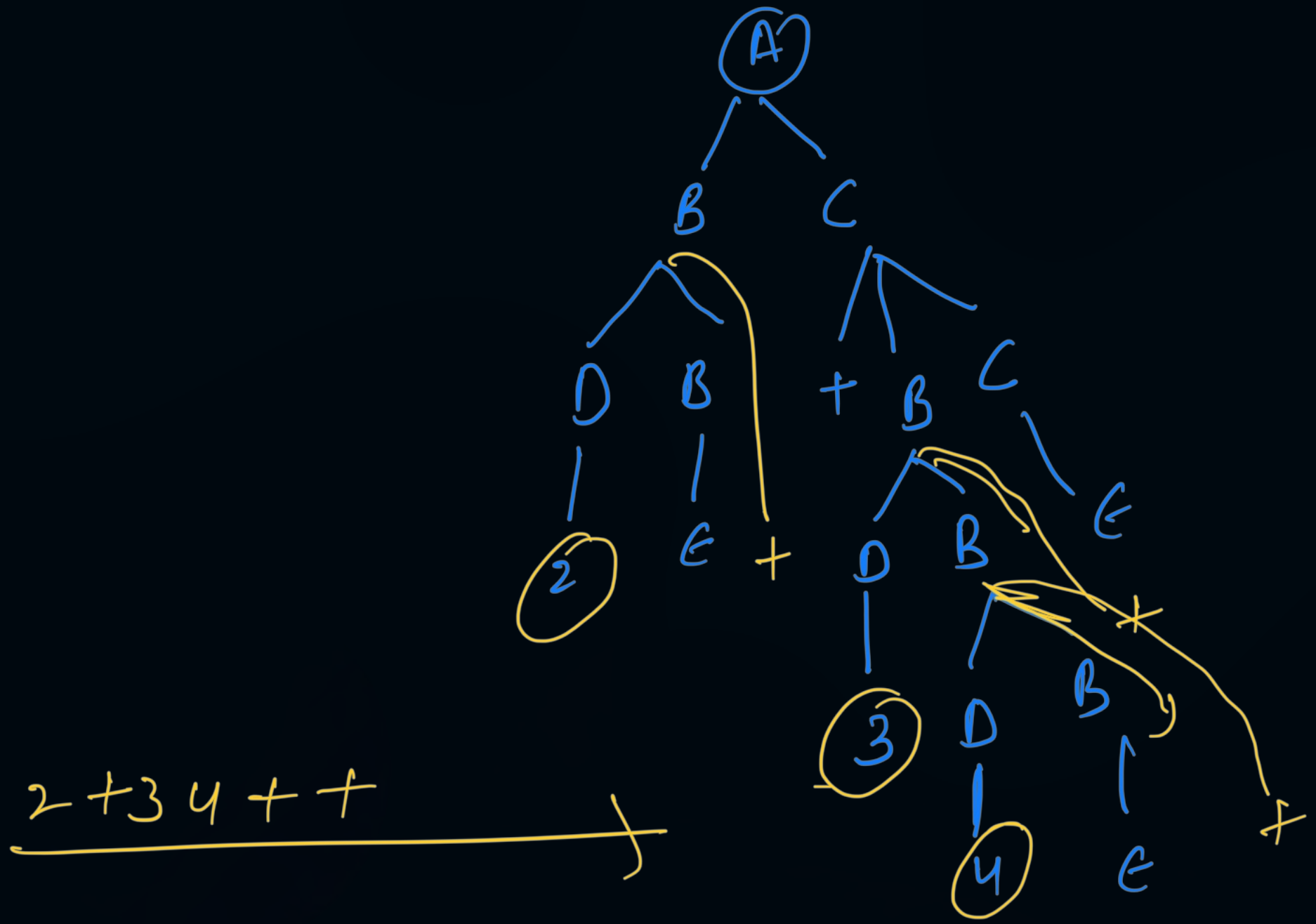
If input is given "2 + 34" then his translation scheme will generate output.

A. 2 + 3 + 4 +

B. + 2 + 3 + 4

C. + + 2 + 34

D. 2 + 34 ++


A
B
C
D
B
+
B
C
2
ϵ
+
D
B
ϵ
3
D
B
*
4
ϵ
+
2 + 3 4 + +

[MCQ]

#Q. ___is performed by attaching rules or algorithms to production in a grammar.

S → aS { Print 3 4 }
S → b

Semantic actions

A. Lexical analysis

B. Execution

C. Syntax directed translation

D. None of these.

[MCQ]

#Q. Consider a translation scheme is given as:

$S \rightarrow S_1 + S_2$ $\{S \cdot val = S_1 \cdot val + S_2 \cdot val\}$

$S \rightarrow S_1 * S_2$ $\{S \cdot val = S_1 \cdot val * S_2 \cdot val\}$

$S \rightarrow id$ $\{S \cdot val = id\}$

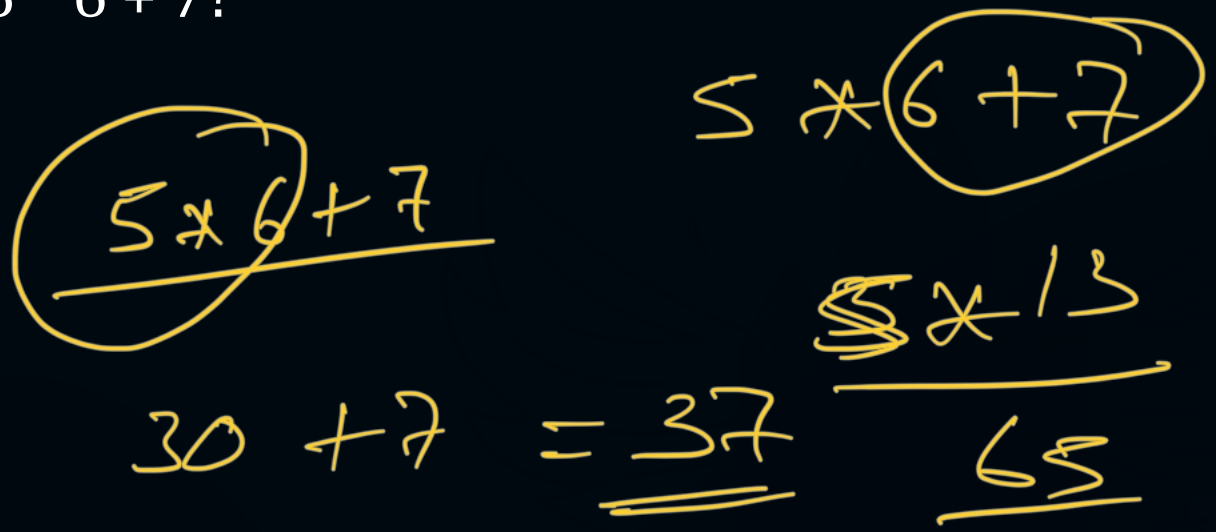
What will be the output for 5 * 6 + 7?

A 18

B 37

C 65

D Cannot be identified because it is ambiguous grammar.

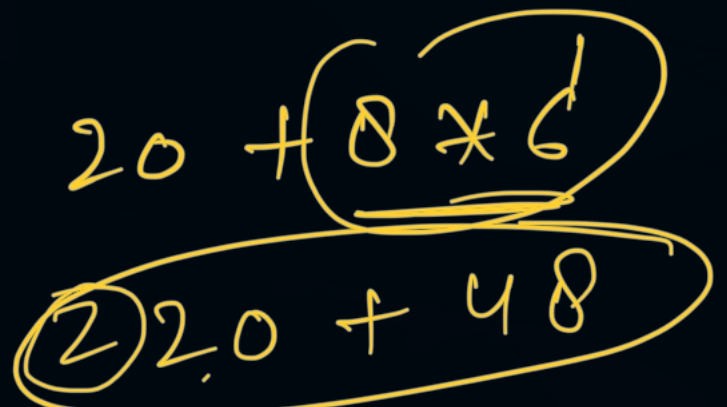


[NAT]

#Q. Consider the following SDT :

$S \rightarrow E$	$\{S \cdot val = E \cdot val\}$
$E \rightarrow E + T$	$\{E \cdot val\} = E_1 \cdot val + T \cdot val\}$
$E \rightarrow T$	$\{E \cdot val = T \cdot val\}$
$T \rightarrow T \times F$	$\{T \cdot val = T_1 \cdot val * F \cdot val\}$
$T \rightarrow F$	$\{T \cdot val = f \cdot val\}$
$F \rightarrow (E)$	$\{F \cdot val = E \cdot val\}$
$F \rightarrow a$	$\{f \cdot val = a\}$

What will be the output of the expression "20 + 8 × 6"



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THANK - YOU